

```

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:1

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:2

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:3

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:4

Stack Menu
1. Push
2. Pop
3. Peek
4. Display
5. Exit
Enter your choice:1
Enter data:5

```

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Q) Write a program to simulate the working of stack using an array with the following:
a) Push b) Pop c) Display. The program should print appropriate messages for stack overflow, stack underflow.

```

#include <stdio.h>
#define N 5

int stack[N];
int top = -1;

void push()
{
    int x;
    printf("Enter data: ");
    scanf("%d", &x);
    if (top == N-1)
    {
        printf("Overflow\n");
    }
    else
    {
        top++;
        stack[top] = x;
    }
}

void pop()
{
    int item;
    if (top == -1)
    {
        printf("Underflow\n");
    }
    else
    {
        item = stack[top];
        printf("%d\n", item);
        top--;
    }
}

```



```

    printf("Underflow\n");
}
else
{
    item = stack[top];
    top--;
    printf("Popped: %d\n", item);
}
}

```

```

void peek()
{
    if (top == -1)
    {
        printf("Underflow\n");
    }
    else
    {
        printf("Top element: %d\n", stack[top]);
    }
}

```

```

int main()
{
    int choice;
    while(1)
    {
        printf("\nStack Menu\n");
        printf("1. Push\n");
        printf("2. Pop\n");
        printf("3. Peek\n");
        printf("4. Exit\n");
    }
}

```



```
printf("Enter your choice:");
scanf("%d", &choice);
```

```
switch(choice)
```

```
{
```

```
case 1:
```

```
    push();
```

```
    break;
```

```
case 2:
```

```
    pop();
```

```
    break;
```

```
case 3:
```

```
    peek();
```

```
    break;
```

```
case 4:
```

```
    return 0;
```

```
default:
```

```
    printf("Invalid choice\n");
```

```
}
```

```
}
```

```
return 0;
```

```
}
```

O/P Stack Menu

1. Push

2. Pop

3. Peek

4. Exit

Enter your choice: 1

Enter data: 10

Stack Menu

1. Push

2. Pop

3. Peek

4. Exit

Enter your choice: 3

Top element: 20

Stack Menu

1. Push

2. Pop

3. Peek

4. Exit

Enter your choice: 2

Popped: 10

Stack Menu

1. Push

2. Pop

3. Peek

4. Exit

Enter your choice: 4

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